

10.3.3 Simple tridiagonal QR algorithm ¶

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fit width



Now, consider the 4×4 tridiagonal matrix

$$\begin{pmatrix} \alpha_{0,0} & \alpha_{0,1} & 0 & 0 \\ \alpha_{1,0} & \alpha_{1,1} & \alpha_{1,2} & 0 \\ 0 & \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{pmatrix}.$$

From $\begin{pmatrix} \alpha_{0,0} \\ \alpha_{1,0} \end{pmatrix}$, one can compute $\gamma_{1,0}$ and $\sigma_{1,0}$ so that

$$\begin{pmatrix} \gamma_{1,0} & -\sigma_{1,0} \\ \sigma_{1,0} & \gamma_{1,0} \end{pmatrix}^T \begin{pmatrix} \alpha_{0,0} \\ \alpha_{1,0} \end{pmatrix} = \begin{pmatrix} \hat{\alpha}_{0,0} \\ 0 \end{pmatrix}.$$

Then

$$\left(\begin{array}{cc|cc} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & 0 \\ \hline 0 & \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{array} \right) = \left(\begin{array}{cc|cc} \gamma_{1,0} & \sigma_{1,0} & 0 & 0 \\ -\sigma_{1,0} & \gamma_{1,0} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \left(\begin{array}{cc|cc} \alpha_{0,0} & \alpha_{0,1} & 0 & 0 \\ \alpha_{1,0} & \alpha_{1,1} & \alpha_{1,2} & 0 \\ \hline 0 & \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{array} \right).$$

Next, from $\begin{pmatrix} \hat{\alpha}_{1,1} \\ \alpha_{2,1} \end{pmatrix}$, one can compute $\gamma_{2,1}$ and $\sigma_{2,1}$ so that

$$\begin{pmatrix} \gamma_{2,1} & -\sigma_{2,1} \\ \sigma_{2,1} & \gamma_{2,1} \end{pmatrix}^T \begin{pmatrix} \hat{\alpha}_{1,1} \\ \alpha_{2,1} \end{pmatrix} = \begin{pmatrix} \hat{\alpha}_{1,1} \\ 0 \end{pmatrix}.$$

Then

$$\left(\begin{array}{cc|cc} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ \hline 0 & 0 & \hat{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{array} \right) = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & \gamma_{2,1} & \sigma_{2,1} & 0 \\ \hline 0 & -\sigma_{2,1} & \gamma_{2,1} & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \left(\begin{array}{cc|cc} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & 0 \\ \hline 0 & \alpha_{2,1} & \alpha_{2,2} & \alpha_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{array} \right)$$

Finally, from $\begin{pmatrix} \hat{\alpha}_{2,2} \\ \alpha_{3,2} \end{pmatrix}$, one can compute $\gamma_{3,2}$ and $\sigma_{3,2}$ so that

$$\begin{pmatrix} \gamma_{3,2} & -\sigma_{3,2} \\ \sigma_{3,2} & \gamma_{3,2} \end{pmatrix}^T \begin{pmatrix} \hat{\alpha}_{2,2} \\ \alpha_{3,2} \end{pmatrix} = \begin{pmatrix} \hat{\alpha}_{2,2} \\ 0 \end{pmatrix}.$$
 Then

$$\left(\begin{array}{cc|cc} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ \hline 0 & 0 & \hat{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & 0 & \hat{\alpha}_{3,3} \end{array} \right) = \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & \gamma_{3,2} & \sigma_{3,2} \\ 0 & 0 & -\sigma_{3,2} & \gamma_{3,2} \end{array} \right) \left(\begin{array}{cc|cc} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ \hline 0 & 0 & \hat{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & \alpha_{3,2} & \alpha_{3,3} \end{array} \right)$$

The matrix Q is the orthogonal matrix that results from multiplying the different Givens' rotations together:

$$Q = \left(\begin{array}{cc|cc} \gamma_{1,0} & -\sigma_{1,0} & 0 & 0 \\ \sigma_{1,0} & \gamma_{1,0} & 0 & 0 \\ \hline 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & \gamma_{2,1} & -\sigma_{2,1} & 0 \\ \hline 0 & \sigma_{2,1} & \gamma_{2,1} & 0 \\ 0 & 0 & 0 & 1 \end{array} \right) \left(\begin{array}{cc|cc} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ \hline 0 & 0 & \gamma_{3,2} & -\sigma_{3,2} \\ 0 & 0 & \sigma_{3,2} & \gamma_{3,2} \end{array} \right). \tag{10.3.1}$$

However, it needs not be explicitly formed, as we exploit next.

The next question is how to compute RQ given the QR factorization of the tridiagonal matrix. We notice that

$$\begin{pmatrix} \hat{\alpha}_{0,0} & \hat{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ 0 & \hat{\alpha}_{1,1} & \hat{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ 0 & 0 & \hat{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & 0 & \hat{\alpha}_{3,3} \end{pmatrix} \begin{pmatrix} \gamma_{1,0} & -\sigma_{1,0} & 0 & 0 \\ \sigma_{1,0} & \gamma_{1,0} & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & \gamma_{2,1} & -\sigma_{2,1} & 0 \\ 0 & \sigma_{2,1} & \gamma_{2,1} & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & \gamma_{3,2} & -\sigma_{3,2} \\ 0 & 0 & \sigma_{3,2} & \gamma_{3,2} \end{pmatrix} \\
\begin{pmatrix} \tilde{\alpha}_{0,0} & \tilde{\alpha}_{0,1} & \hat{\alpha}_{0,2} & 0 \\ \tilde{\alpha}_{1,0} & \tilde{\alpha}_{1,1} & \hat{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ 0 & 0 & \hat{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & 0 & \hat{\alpha}_{3,3} \end{pmatrix} \\
\begin{pmatrix} \tilde{\alpha}_{0,0} & \tilde{\alpha}_{0,1} & \tilde{\alpha}_{0,2} & 0 \\ \tilde{\alpha}_{1,0} & \tilde{\alpha}_{1,1} & \tilde{\alpha}_{1,2} & \hat{\alpha}_{1,3} \\ 0 & \tilde{\alpha}_{2,1} & \tilde{\alpha}_{2,2} & \hat{\alpha}_{2,3} \\ 0 & 0 & 0 & \hat{\alpha}_{3,3} \end{pmatrix} \\
\begin{pmatrix} \tilde{\alpha}_{0,0} & \tilde{\alpha}_{0,1} & \tilde{\alpha}_{0,2} & 0 \\ \tilde{\alpha}_{1,0} & \tilde{\alpha}_{1,1} & \tilde{\alpha}_{1,2} & \tilde{\alpha}_{1,3} \\ 0 & \tilde{\alpha}_{2,1} & \tilde{\alpha}_{2,2} & \tilde{\alpha}_{2,3} \\ 0 & 0 & \tilde{\alpha}_{3,2} & \tilde{\alpha}_{3,3} \end{pmatrix}.
\end{pmatrix}$$

A symmetry argument can be used to motivate that $\tilde{\alpha}_{0,2} = \tilde{\alpha}_{1,3} = 0$ (which is why they appear in gray, if you look carefully). This also explains why none of the elements above the first superdiagonal become nonzero.

Remark 10.3.3.1. An important observation is that if A is tridiagonal, then $A \rightarrow QR$ (QR factorization) followed by $A := RQ$ again yields a tridiagonal matrix. In other words, any QR algorithm previously discussed (simple, shifted, with deflation) when started with a tridiagonal matrix will generate a succession of tridiagonal matrices.